Lakshimi Raman S

Software Engineer, Infrastructure

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Software Engineer, Infrastructure on Zoho's SaaS Framework Team (15 engineers) — operating 900+ PostgreSQL and MySQL clusters across 60+ data centers for 50+ applications serving 80M+ tenants. Builds cluster orchestration, automated failover, zero-downtime upgrades, custom WAL decoders, multi-engine backup/DR frameworks, and connection-pool failover. Strong in distributed systems, sharding, data replication, high availability, and database internals. Core: Go, C.

SKILLS

Languages: Go, C, C++, Java, Python, JavaScript, SQL, Bash

Databases: PostgreSQL, MySQL, ClickHouse, etcd, Tarantool, DGraph, TiKV, SQLite; familiar with Neo4j, JanusGraph, SurrealDB

DB Internals: WAL, logical/physical replication, custom WAL decoders, PostgreSQL hooks, B-Tree, MVCC, Foreign Data Wrappers

Dist. Systems: etcd/Raft, witness-quorum failover, blue-green deploy, sharding, consistent hashing

Infra: ProxySQL, PgBouncer, Docker, Linux, AWS, Prometheus, Grafana, systemd

Ops: Backup/restore, PITR, DR, capacity planning, query tuning, on-call incident response

EXPERIENCE

Technical Trainee

Chennai, India | Jun 2024 – May 2025

Zoho Corporation — SaaS Framework Team Pre-hire internship | database infrastructure engineering

- **Custom WAL decoder:** Extended PostgreSQL logical replication with query-pipeline hook and custom WAL decoder for DDL propagation; 2.5× faster than prior ORM approach, eliminated schema-drift bugs.
- **Zero-downtime upgrades:** Re-architected blue-green PostgreSQL major-version upgrades from 6 spare nodes to 2; eliminates ~40 high-spec nodes (256 GB RAM / 10 TB) per upgrade cycle.

Software Engineer, Infrastructure

Chennai, India | Jun 2025 – Present

Zoho Corporation — SaaS Framework Team 15 engineers | 900+ clusters | 60+ DCs | 80M+ tenants | 50+ products

- **Failover:** Reduced cluster failover 1–2 min to ~5 sec via witness-service quorum in Go; eliminated manual split-brain resolution across 900+ clusters.
- **Backup framework:** Topology-agnostic system (linear, quorum, quorum-with-roles, cluster) handling ~2 TB/week across 6 engines (PostgreSQL, MySQL, ClickHouse, etcd, Tarantool, Prometheus); validated end-to-end on ClickHouse; pull model avoids credential exposure; hooks API for app teams; GPG encryption + restore.
- **ProxySQL & connection pool:** Managed ProxySQL across 100+ clusters (read/write routing cut primary load ~40%); built dual-datasource failover in the pool enabling VLAN/port migrations across 500+ app servers with zero restarts.
- On-call across 900+ clusters (replication lag, deadlocks, MVCC bloat, VLAN issues); attended PGConf India 2026 and connected with ClickHouse co-founder to apply PostgreSQL scaling patterns to multi-tenant infra.

PROJECTS

Cluster Management System | *Go, etcd*

- Orchestrates 900+ MySQL/PostgreSQL clusters: witness-quorum, auto-failover, VIP migration, rolling upgrades, resumable workflows, ProxySQL/PgBouncer integration.

Hot/Cold Partitioning POC | *PostgreSQL FDW, Parquet*

- 600 GB table (500 GB cold) offloaded via PostgreSQL FDW to Parquet; 80% cold-data footprint reduction, ~50% faster hot queries from freed disk and cleared index bloat.

DGraph Translational Layer | *Go, DGraph — in dev*

- RDF-to-property-graph translation via relationship-as-node model; exposes DGraph to 50+ app teams without RDF/DQL knowledge; edge properties as first-class nodes.

Internal-workings | *C, Go*

- Paged disk manager (bitmap, mmap); 200-line C allocator (sbrk/mmap/free-list); linear-probing hash map; TCP socket server — all from scratch.
- github.com/LakshimiRam-073/Internal-workings

Open Source — Dgraph (dgraph-io) | *Go, Raft*

- Contributed configurable Raft `election-tick` to the consensus layer (Alpha + Zero nodes) for leader-election control across heterogeneous network topologies; added fail-fast validation after maintainer review. Open PR #9725, under review.

EDUCATION

B.E. — Electronics & Communication Engineering

Sri Krishna College of Engineering and Technology (SKCET), Coimbatore

Oct 2021 – May 2025 | CGPA: 8.44 / 10

ACCOMPLISHMENTS

- **Google Foobar — Level 4:** Invite-only algorithmic challenge; top ~5% globally.
- **Smart India Hackathon — National Finalist** (100,000+ participants).
- **BIT Virupaksha ML Hackathon — Finalist.**